

Chapter 2

POWER PLANTS

2.1 UTILISATION OF ENERGY

This chapter deals with the study of utilisation of various sources of energy for the generation of power in power plants, i.e., layout of various types of power plants, their major components and functions, including relative advantages and disadvantages of each.

2.2 STEAM (THERMAL) POWER PLANT

Working Principle

A steam power plant, also known as thermal power plant, is using steam as working fluid. Steam is produced in a boiler using coal as fuel and is used to drive the prime mover, namely, the steam turbine. In the steam turbine, heat energy is converted into mechanical energy which is used for generating electric power. Generator is an electro-magnetic device which makes the power available in the form of electrical energy.

Layout of Steam Power Plant : (Fig. 2.1)

The layout of the steam power plant is shown in Fig. 2.1. It consists of four main circuits. These are : (i) Coal and Ash circuit, (ii) Air and Flue Gas circuit, (iii) Water and Steam circuit and (iv) Cooling water circuit

i) Coal and Ash Circuit :

Coal from the storage yard is transferred to the boiler furnace by means of coal handling equipment like belt conveyor, bucket elevator, etc. Ash resulting from the combustion of coal in the boiler furnace collects at the back of the boiler and is removed to the ash storage yard through the ash handling equipment.

Ash Disposal : The Indian coal contains 30 to 40% ash. A power plant of 100 MW capacity produces 20 to 25 tonnes of hot ash per hour. Hence sufficient space near the power plant is essential to dispose such large quantities of ash.

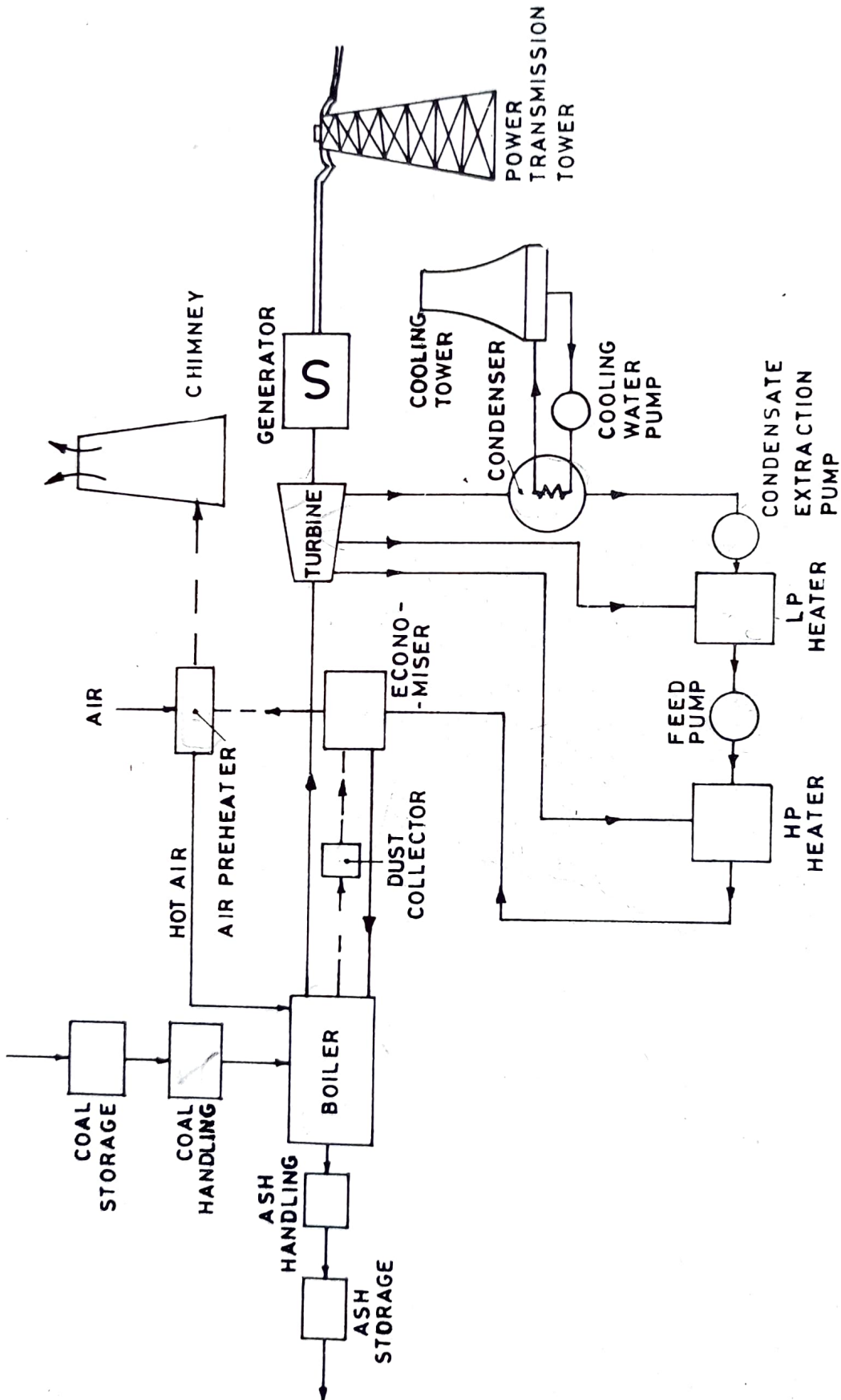


Fig. 2.1 Layout of Steam (Thermal) Power Plant

ii) Air and Flue Gas Circuit :

Air is taken in from the atmosphere to the air preheater. Air is heated in the air preheater by the heat of the flue gas which is passing to the chimney. The hot air is supplied to the furnace of the boiler.

The flue gases after combustion in the furnace, pass around the boiler tubes. The flue gases then pass through a dust collector, economiser and pre-heater before being exhausted to the atmosphere through the chimney. By this method the heat of the flue gases which would have been wasted otherwise is used effectively. Thus the overall efficiency of the plant is improved.

Air Pollution : The pollution of the surrounding atmosphere is caused by the emission of objectionable gases and dust through the chimney. The air pollution and smoke cause nuisance to people surrounding the plant.

iii) Feed Water and Steam Circuit :

The steam generated in the boiler passes through super-heater and is supplied to the steam turbine. Work is done by the expansion of steam in the turbine and the pressure of steam is reduced. The expanded steam then passes to the condenser, where it is condensed.

The condensate leaving the condenser is first heated in a L.P. water heater by using the steam taken from the low pressure extraction point of the turbine. Again steam taken from the high pressure extraction point of the turbine is used for heating the feed water in the H.P. water heater. The hot feed water is passing through the economiser, where it is further heated by means of flue gases. The feed water which is sufficiently heated by the feed water heaters and economiser is then fed into the boiler.

iv) Cooling Water Circuit :

Abundant quantity of water is required for condensing the steam in the condenser. Water circulating through the condenser may be taken from various sources such as river or lake, provided adequate water supply is available from the river or lake throughout the year.

If adequate quantity of water is not available at the plant site, the hot water from the condenser is cooled in cooling tower or cooling ponds and circulated again.

Advantages of thermal power plants

1. Initial cost is low compared with hydro-plant.
2. The power plant can be located near load centre, so the transmission cost and transmission losses are considerably reduced.

3. The generation of power is not dependent on the nature's mercy like hydro-plant.
4. The construction and commissioning of thermal plant requires less period of time than a hydro-plant.

Disadvantages

1. The fuel (coal) needed may be exhausted by gradual use.
2. Efficiency of the plant decreases with time.
3. It cannot be used as peak load plant, as its part load efficiency decreases very rapidly with decreasing load.
4. Transportation of fuel is one of the major difficulties for the plants located away from coal fields.
5. Power generation cost is considerably high compared with hydro-plant.
6. Air pollution causing smoke in the surrounding atmosphere is a major problem with thermal plant.
7. Life of the plant is hardly 3 to 4 decades compared with hydro-plant (1 to 2 centuries).

2.3 HYDRO-ELECTRIC POWER PLANT

Working Principle

Hydro-electric power plant utilises the potential energy of water stored in a dam built across the river. The potential energy of water is used to run water turbine to which the electric generator is coupled. The mechanical energy available at the shaft of the turbine is converted into electrical energy by means of the generator.

General arrangement of a Hydro-Electric Power Plant

Fig. 2.2 shows the schematic representation of a hydro-electric power plant.

i) Water Reservoir :

Continuous availability of water is the basic necessity for a hydro-electric plant. Water collected from catchment area during rainy season is stored in the reservoir. Water surface in the storage reservoir is known as head race.

ii) Dam :

The function of a dam is to increase the height of water level behind it, which ultimately increases the reservoir capacity. The dam also helps to increase the working head of the power plant.

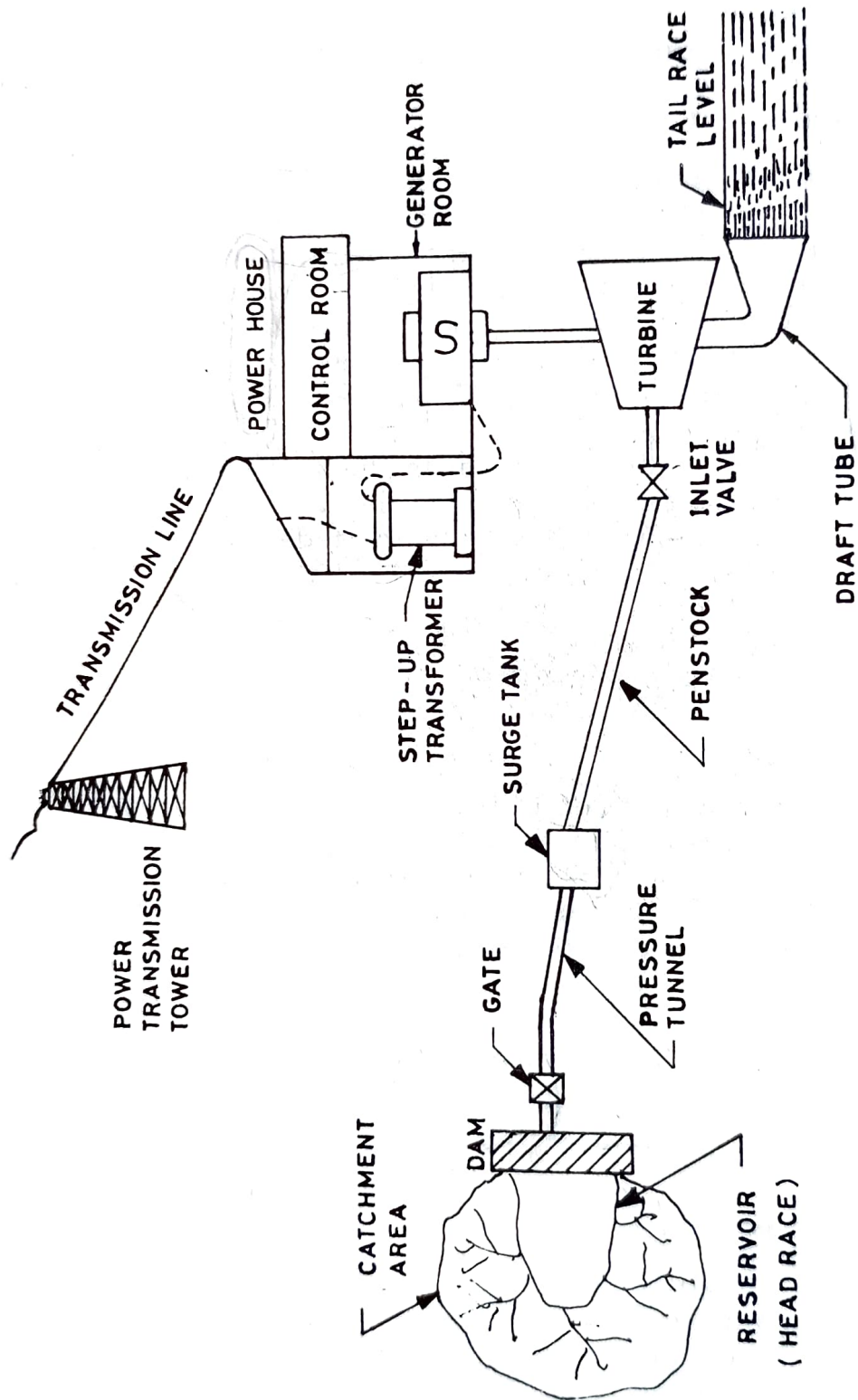


Fig. 2.2 Layout of Hydro-Electric Power Plant

iii) Spillway :

Water after a certain level in the reservoir overflows through spillway without allowing the increase in water level in the reservoir during rainy season.

iv) Pressure Tunnel :

It carries water from the reservoir to surge tank.

v) Penstock :

Water from surge tank is taken to the turbine by means of penstocks, made up of reinforced concrete pipes or steel.

vi) Surge Tank

There is sudden increase of pressure in the penstock due to sudden backflow of water, as load on the turbine is reduced. The sudden rise of pressure in the penstock is known as water hammer. The surge tank is introduced between the dam and the power house to keep in reducing the sudden rise of pressure in the penstock. Otherwise penstock will be damaged by the water hammer.

vii) Water Turbine :

Water through the penstock enters into the turbine through an inlet valve. Prime movers which are in common use are Pelton turbine, Francis turbine and Kaplan turbine. The potential energy of water entering the turbine is converted to mechanical energy. The mechanical energy available at the turbine shaft is used to run the electric generator. The water is then discharged through the draft tube.

viii) Draft Tube :

It is connected to the outlet of the turbine. It allows the turbine to be placed over tail race level.

ix) Tail Race :

Tail race is a water way to lead the water discharged from the turbine to the river. The water held in the tail race is called tail race water level.

x) Step-up Transformer :

Its function is to raise the voltage generated at the generator terminal before transmitting the power to consumers.

xi) Power House :

The power house accommodates the turbine, generator, transformer and control room.

Classification of Hydro-Power Plants

Hydro-plants are classified according to the head of water under which they work.

When the operating head of water exceeds 70 metres, the plant is known as "high head power plant". Pelton turbine is used as prime mover in such power plants.

When the head of water ranges from 15 to 70 metres then the power plant is known as "medium head plant". It uses Francis turbine.

When the head is less than 15 metres the plant is named as "low head plant". It uses Francis or Kaplan turbine as prime mover.

Advantages of Hydro-Electric Power Plants

1. Water is a renewable source of energy. Water which is the operating fluid, is neither consumed nor converted into something else.

2. Water is the cheapest source of energy because it exists as a free gift of nature. The fuels needed for the thermal, diesel and nuclear plants are exhaustible and expensive.

3. There is no ash disposal problem as in the case of thermal power plant.

4. Hydro-plant does not pose the problem of air pollution as in the case of thermal plant or radiation hazards as in the case of nuclear plant.

5. Variable loads do not affect the efficiency in the case of a hydro-plant.

6. Life of hydro-plant is very long (1 or 2 centuries) compared with thermal plant (3 to 4 decades). This is because the hydro-plants operate at atmospheric temperature, whereas thermal plants operate at very high temperatures (about 500 to 800°C).

7. Hydro-plant provides additional benefits like irrigation, flood control, fishery and recreation.

8. The water storage of hydro-plant can also be used for domestic water supply.

9. Auxiliaries needed for the hydro-plant are less compared to thermal plant of equal capacity.

10. It requires less supervising staff.
11. Maintenance cost is low.

Disadvantages

1. Hydro-plants are generally situated away from the load centres. Hence long transmission lines are required for delivery of power. This increases the cost of transmission lines and also transmission losses. But a thermal plant can be located near the load centre, thereby the transmission cost and transmission losses are considerably reduced.
2. The power produced by hydro-plant depends upon the quantity of water which in turn is dependent upon the rainfall. The dry year affects the hydro-power generation considerably.
3. Initial cost of the plant is high.
4. Erection of hydro-plant (construction of dam etc) usually takes long period of time.

2.4 NUCLEAR POWER PLANT

Working Principle

A nuclear power plant differs from a conventional steam power plant only in the steam generating part. There is no change in the turbo-alternator and the condensing system.

The nuclear fuel which is at present in commercial use is Uranium. Scientists say that 1 kg of U^{235} can produce as much energy as can be produced by burning 4500 tonnes of high grade coal.

Nuclear Fission : Refer Fig. 2.3 .

Uranium exists in the isotopic form of U^{235} which is unstable. When a neutron enters the nucleus of U^{235} , the nucleus splits into two equal fragments and also releases 2.5 fast moving neutrons with a velocity of 1.5×10^7 metres/sec producing a large amount of energy, nearly 200 million electron-volts. This is called "nuclear fission".

Chain Reaction :

The neutrons released during the fission can be made to fission other nuclei of U^{235} causing a "chain reaction". A chain reaction produces enormous amount of heat, which is used to produce steam.

The chain reaction under uncontrolled conditions can release extremely large amounts of energy causing "atomic explosion".

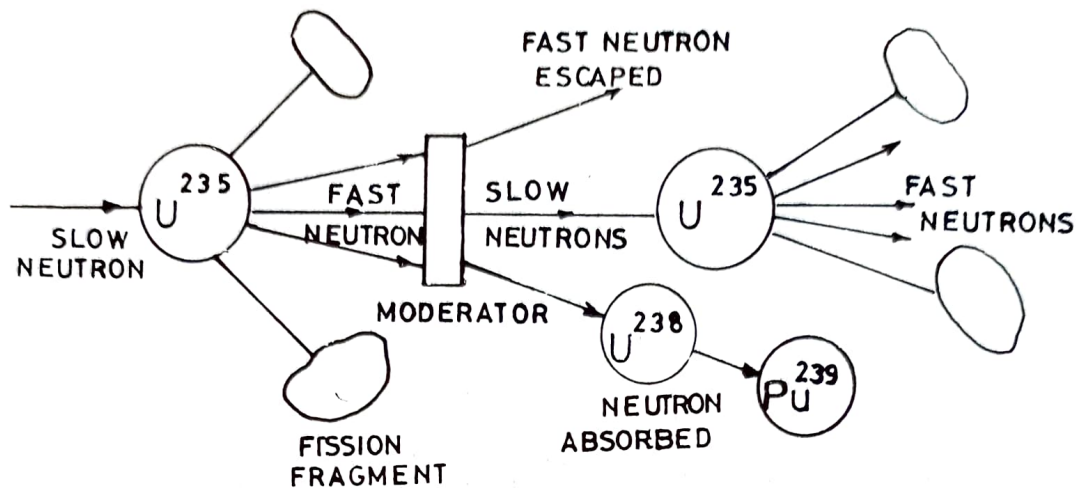


Fig. 2.3 Nuclear Fission

Energy liberated in chain reaction, according to Einstein Law, is $E = mv^2$, where E = energy liberated, m = mass in grams, v = speed of light = 3×10^{10} cm/sec.

Out of 2.5 neutrons released in fission of each nuclei of U^{235} , one neutron is used to sustain the chain reaction, 0.9 neutron is converted into fissionable material Pu^{239} and 0.6 neutron is absorbed by control rod and coolant moderator.

Function of the moderator is to reduce the energy of neutrons evolved during fission in order to maintain the chain reaction. The moderators which are commonly used are ordinary water and heavy water.

Main components of a Nuclear Power Plant : (Fig. 2.4)

i) Nuclear Reactor :

A nuclear reactor may be regarded as a substitute for the boiler fire box of a steam power plant. Heat is produced in the reactor due to nuclear fission of the fuel U^{235} . The heat liberated in the reactor is taken up by the coolant circulating through the core. Hot coolant leaves the reactor at top and flows into the steam generator (boiler).

Radiation Hazards and Shieldings :

The reactor is a source of intense radio-activity. These radiations are very harmful to human life. It requires strong control to ensure that this radio-activity is not released into the atmosphere to avoid atmospheric pollution. A thick concrete shielding and a pressure vessel are provided to prevent the escape of these radiations to atmosphere.

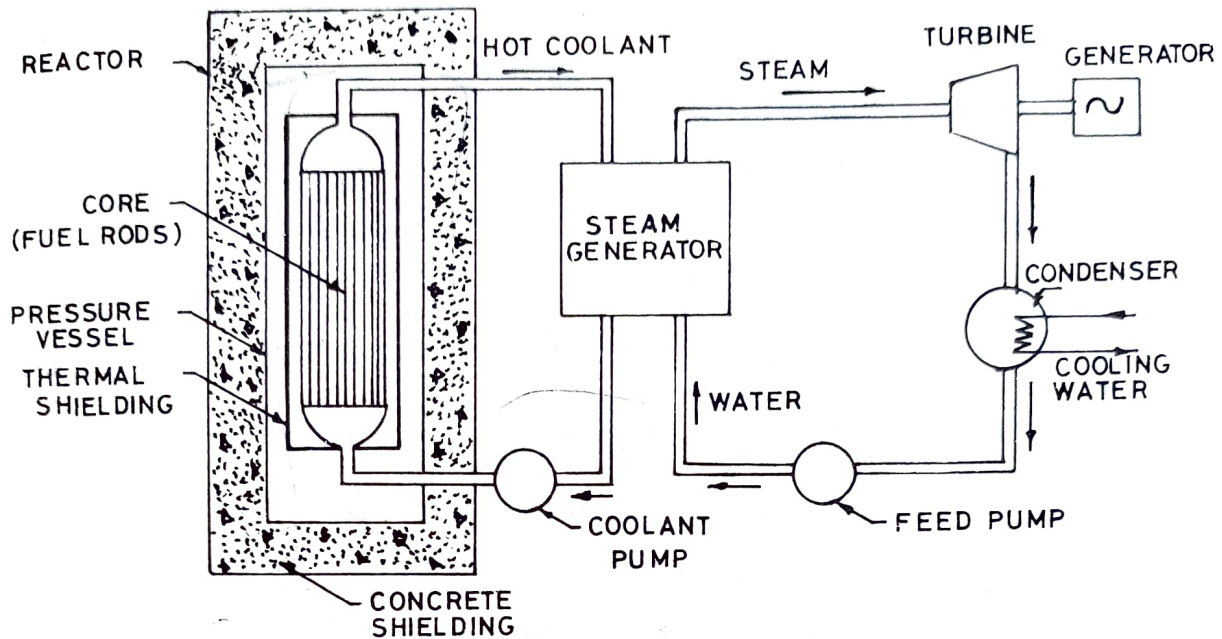


Fig. 2.4 Nuclear Power Plant

Types of Reactors :

1. Pressurised Water Reactor (P.W.R.) shown in Fig. 2.4.
2. Boiling Water Reactor (B.W.R.).
3. Heavy Water-Cooled Reactor.

ii) Steam Generator :

The steam generator is fed with feed water which is converted into steam by the heat of the hot coolant. The purpose of the coolant is to transfer the heat generated in the reactor core and use it for steam generation. Ordinary water or heavy water is a common coolant.

iii) Turbine :

The steam produced in the steam generator is passed to the turbine and work is done by the expansion of steam in the turbine.

iv) Coolant Pump and Feed Pump :

The steam from the turbine flows to the condenser where cooling water is circulated. Coolant pump and feed pump are provided to maintain the flow of coolant and feed water respectively.

Advantages of Nuclear Power Plant

1. It can be easily adopted where water and coal resources are not available.
2. The nuclear power plant requires very small quantity of fuel. Hence fuel transportation cost is less.
3. Space requirement is less compared to other power plants of equal capacity.
4. It is not affected by adverse weather conditions.
5. Fuel storage facilities are not needed as in the case of the thermal power plant.
6. Nuclear power plants will conserve the fossil fuels (coal, petroleum) for other energy needs.
7. Number of workmen required at nuclear plant is far less than thermal plant.
8. It does not require large quantity of water.

Disadvantages

1. Radio-active wastes, if not disposed carefully, have adverse effect on the health of workmen and the population surrounding the plant.
2. It is not suited for varying load conditions.
3. It requires well-trained personnel.
4. It requires high initial cost compared to hydro or thermal power plants.

Clean atomic power by Nuclear Fusion

Nuclear Fusion : Nuclear fusion technology is based on producing energy by uniting atoms, while nuclear fission achieves this by splitting atoms. Nuclear fusion is the process by which light elements are combined to form a single element with a release of energy. The two light elements fuse into a heavier element. Hydrogen nuclei are made to fuse to form Helium nucleus. The difference in mass is converted into energy. Fusion reactors do not give out harmful radiation.

A laboratory breakthrough has been achieved in U.S.A., in March 1989 in the area of nuclear fusion for the production of clean atomic power without creating any environmental problem.

Scientists in the near future will achieve a factory-level technology for commercial production of power through nuclear fusion. It could thus provide a key to the production of nuclear energy without the accompanying threat of accumulation of fissionable radio-active material that could otherwise be used for making nuclear weapons.

2.7 NON-CONVENTIONAL POWER GENERATING SYSTEMS USING RENEWABLE SOURCES OF ENERGY

The non-renewable conventional sources of energy like coal, diesel, natural gas, etc., will be exhausted by the middle of the next century, if the present rate of power increase and population increase continue. As these non-renewable sources are consumed, the mankind must turn its attention to longer-term and permanent type of energy resources. The Scientists and Engineers all over the world are in search of new non-conventional sources of energy for the last few decades. These sources, viz., solar, wind, tidal, etc., are considered renewable and non-exhaustible as these are considered as perpetual sources of energy. Solar energy and wind energy (to certain extent) show promise of becoming dependable renewable energy sources.

2.8 SOLAR POWER PLANT

Basic Principle of Solar Energy

The sun gives out 3.7×10^{20} MW of energy into space, out of which earth intercepts only 1.7×10^{14} MW. Solar radiation is reduced in intensity in the atmosphere by clouds, dust, haze, and fog. The energy emitted by the sun in three minutes is equivalent to the world energy consumption during a year. Most of the energy we receive from the sun comes in the form of light, a short wave radiation, not all of which is visible to the human eye. When this radiation strikes a solid or liquid, it is absorbed and transformed into heat energy.

The heat energy is collected in a fluid, generally water. This heat energy can be used in a solar heater or drier or can be used to run an engine. Flat plate collector or parabolic concentrated type collector is used to collect and concentrate the solar energy and increase the temperature of the working fluid.

Applications of solar energy enjoying most success today are:

1. Solar engines for water pumping.
2. Solar water heaters.
3. Solar cookers.
4. Solar driers.
5. Solar furnaces.
6. Photo-voltaic conversion (solar cells).
7. Solar power generation.

Arrangement of a Solar Power Plant :

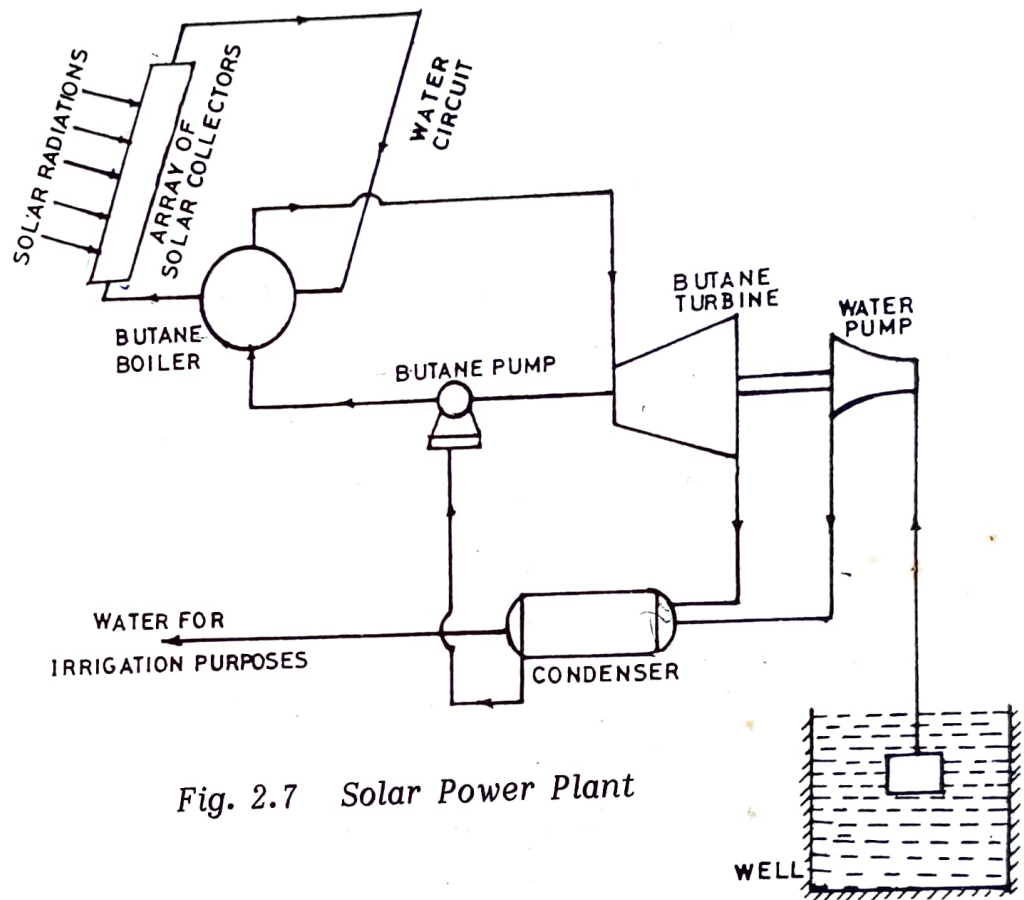


Fig. 2.7 Solar Power Plant

Fig. 2.7 shows a solar power plant with a low temperature solar engine using heated water from flat plate solar collector and Butane as the working fluid. This was developed in France for lift irrigation.

i) Solar Collectors :

Flat plate collector : (Fig. 2.8) In a flat plate collector, the radiation energy of the sun falls on a flat surface coated with black paint having high absorbing capacity. It is placed facing the general direction of the sun. The materials used for the plate may be copper, steel or aluminium. The thickness of the plate is 1 to 2mm. Tubing of copper is provided in thermal contact with the plate.

Heat is transferred from the absorber plate to water which is circulated in the copper tubes through the flat plate collector.

Thermal insulation is provided behind the absorber plate to prevent heat losses from the rear surface. Insulation material is generally fibre glass or mineral wool. The front cover is made up of glass and it is transparent to the in-coming solar radiation.

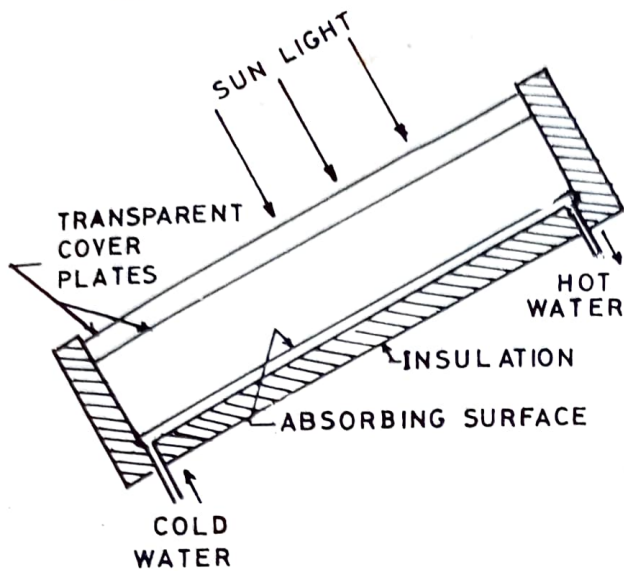


Fig. 2.8 Flat Plate Collector

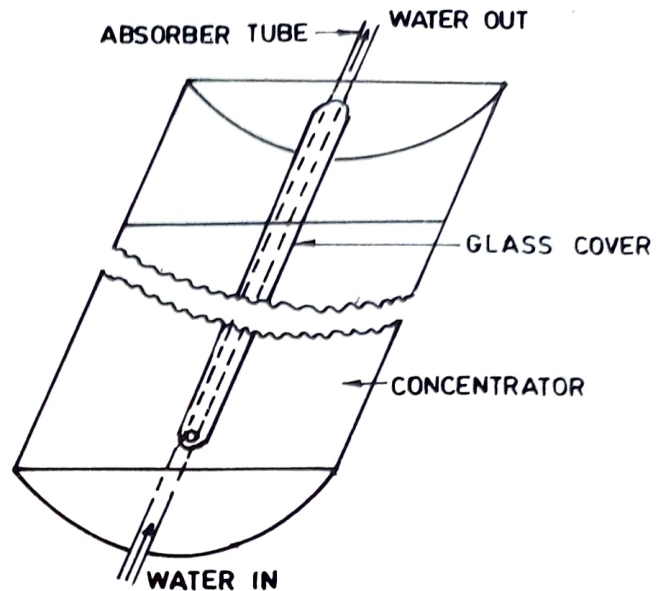


Fig. 2.9 Cylindrical Parabolic Concentrator Collector

Cylindrical parabolic concentrator collector; (Fig. 2.9) : Concentrator collectors are of reflecting type utilising mirrors. The reflecting surface may be a parabolic mirror. The solar energy falling on the collector surface is reflected and focussed along a line where the absorber tube is located. As large quantity of energy falling on the collector surface is collected over a small surface, the temperature of the absorber fluid is very much higher than in flat plate collector.

While flat plate collectors may be used to heat water upto 80°C (low temperature), the concentrating type of collectors are designed to heat water to medium and high temperature ranges.

i) *Butane boiler :*

The water heated in flat plate solar collector to 80°C is used for boiling Butane at high pressure in the Butane boiler. Boiling point of Butane is about 50°C .

iii) *Turbine :*

The Butane vapour generated at high pressure in the boiler is used to run the vapour turbine which drives the electrical generator.

The vapour coming out of the turbine at low pressure is condensed in a condenser using water. The condensed liquid Butane is fed back to the Butane boiler using feed pump.

Tower concept for power generation : (Fig. 2.10)

The tower concept consists of an array of plane mirrors or heliostats which are individually controlled to reflect radiations from the sun into a boiler mounted on a 500 metres high tower.

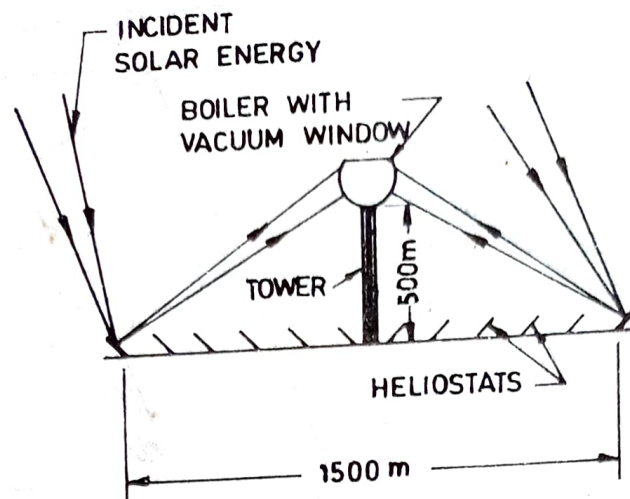


Fig. 2.10 Tower Concept for Power generation

Steam is generated in the boiler, which may attain a temperature upto 2000°K .

Electricity is generated by passing steam through the turbine coupled to a generator. A 50 kW plant based on this concept has been built and successfully operated in Italy.

Advantages

1. Sun is essentially an infinite source of energy. Therefore solar energy is a very large inexhaustible and renewable source of energy and is freely available all over the world.
2. It is environmentally very clean and is hence pollution-free.
3. It is a dependable energy source without new requirements of a highly technical and specialised nature for its wide spread utilization.
4. It is the best alternative for the rapid depletion of fossil fuels.

Disadvantages

1. It is available in a dilute form and is at a low potential. The intensity of solar energy on sunny day in India is about 1.1 kW/square metre area. Hence very large collecting areas are required.
2. Also the dilute and diffused nature of the solar energy needs large land area for the power plant ; for instance, about 30 square kilometers area is required for a solar power station to replace a nuclear plant on a 1 square kilometer site. Hence capital cost is more for the solar plant.
3. Solar energy is not available at night or during cloudy or rainy days.

2.9 WIND POWER GENERATION

The primitive man had to use his muscle power to carry out different works in his attempt of obtaining the necessities of life. The demand for more power and the scarcity of man power and the ready availability of natural resources like wind, water, etc., might have formed the reason why the ancient man turned his attention to the utilisation of these resources as sources of power; thus originated the early prime movers namely wind mill and water wheel.

Principle of Wind Mill

[Wind flow is created as an effect of solar heat. Winds are caused due to the absorption of solar energy on the earth surface and the rotation of earth about its own axis and around the sun. Because of this, alternate heating and cooling occurs and difference in pressure is obtained and the air movement is caused.]

The flowing wind which has kinetic energy is used to rotate the wind turbine which is also known as wind mill. Although wind mills have been used for more than a dozen centuries for pumping water and grinding grains, interest in large scale power generation had developed over the past 50 years.

The earth's atmosphere is thus a marvellous solar driven heat engine. It is estimated that roughly 10 millions MW of energy is continuously available in the earth's winds.

Types of Wind Mills

Wind mills are classified as

- i) Horizontal axis type and
 - ii) Vertical axis type,
- depending on their axis of rotation.

Horizontal axis wind mills are further classified as single-bladed, double-bladed and multi-bladed types.

i) Horizontal axis wind mill with double blade rotor: (Fig. 2.11)

In the horizontal axis single blade type, the blade is of propeller type with counter weight arrangement. The double blade type gives a better performance than the single blade type.

In the double-blade type, the blade has thick cross section of an aerofoil. At the tip of the blade, the velocity is about six times the wind velocity. The blade is set at right angles to the direction of the wind. Ideally the blades should be twisted, but because of construction difficulties this is not always achieved.

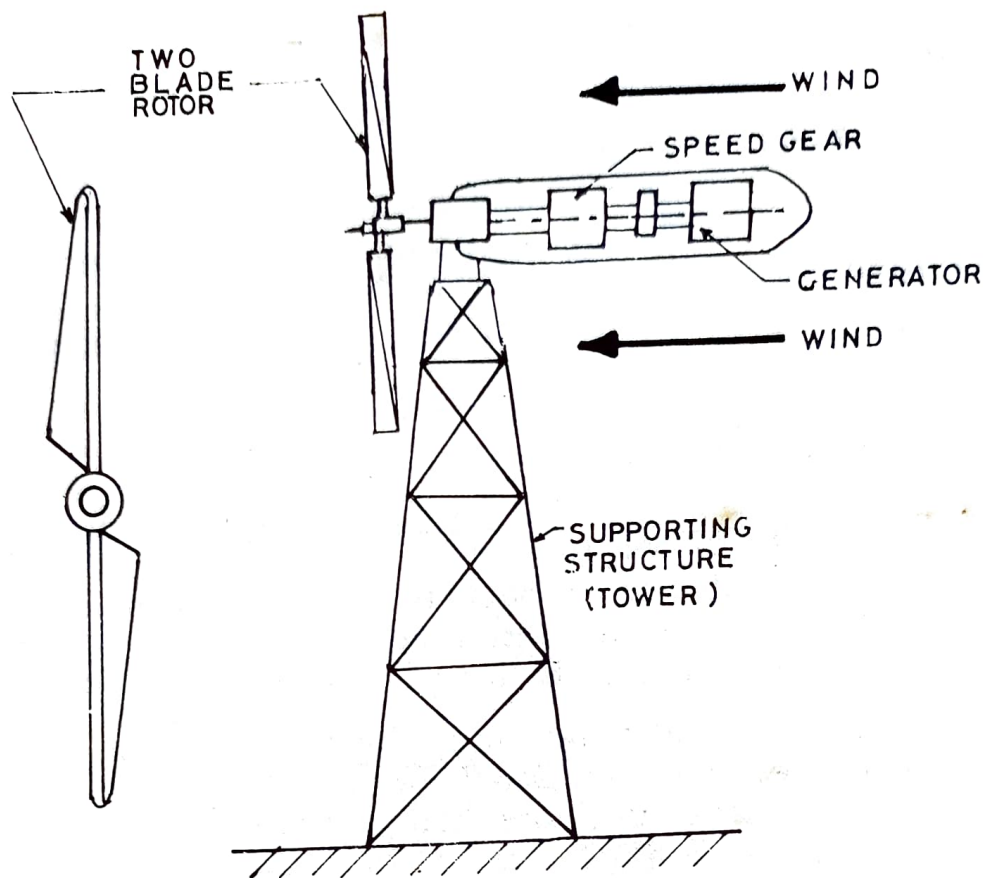


Fig. 2.11 Horizontal Axis Windmill with Double Blade Rotor

The blade rotor drives a generator through a gear box. It is mounted on a bed plate attached on the top of the tower.

Blade material : The blades are made from aluminium or sheet metal. Not a single large wind turbine-generator with metal blades has operated for longer than one year without a blade failure. The suitable material required is a major problem in developing wind mills of higher power generation capacity.

With rotor, the tower is also subjected to the wind loads which may cause serious damage. Hence the structure of the tower should be so designed to withstand the wind load.

The best sites for wind energy are found off-shore and sea coast. The second best sites are in mountains. The lowest level of wind energy is found in plains, where values are generally three or four times lower than that at the coast.

Multiblade Rotor : It is shown in Fig. 2.12. Rotors with more than two blades will produce high power.

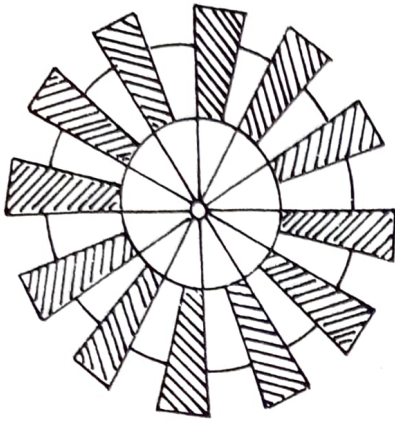


Fig. 2.12 Multiblade Rotor

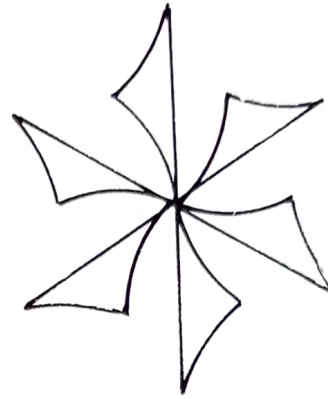


Fig. 2.13 Sail type blades

Sail type blades : (Fig. 2.13)

It is of recent origin. The blade surface is made from cloth, nylon or plastics arranged as sail wings. There is also variation in the number of sails used.

The horizontal axis types generally have better performance than vertical types. They are mainly used for power generation and pumping water. The biggest wind mill erected for power generation is of 2500 kW capacity in U.S.A.

Advantages

1. The wind energy is a renewable source of energy. It is free and inexhaustible.
2. The power requirements for irrigation, lighting and small industrial units can be fulfilled with the use of wind energy. Power generation on large scale using wind energy is not yet so successful, but the small wind mills will play a vital role in the present condition of power shortage.
3. It does not need transportation.
4. Like all forms of solar energy, wind power systems are non-polluting.

Disadvantages

1. Wind velocity is fluctuating which makes the complications in designing a wind power plant.
2. Some form of storage of wind energy is essential to maintain a constant supply of power.

3. The wind is a very hazardous, treacherous and unpredictable source of energy. Blowing in strong gusts from varying directions causing hurricanes, the wind may smash the whole plant within no time. To avoid this, special and costly designs and controls are always required.

4. It is considered suitable and economical to generate power on a small scale of the order of 2 MW.

5. Power production cost with the present technology available may not compete with the conventional power generating systems. This is because, 1000s of units are required to provide the power output of one fossil-fueled plant.

6. Wind energy systems are noisy in operation. A large unit can be heard many kilometers away.

2.10 TIDAL POWER GENERATION : (Fig. 2.14)

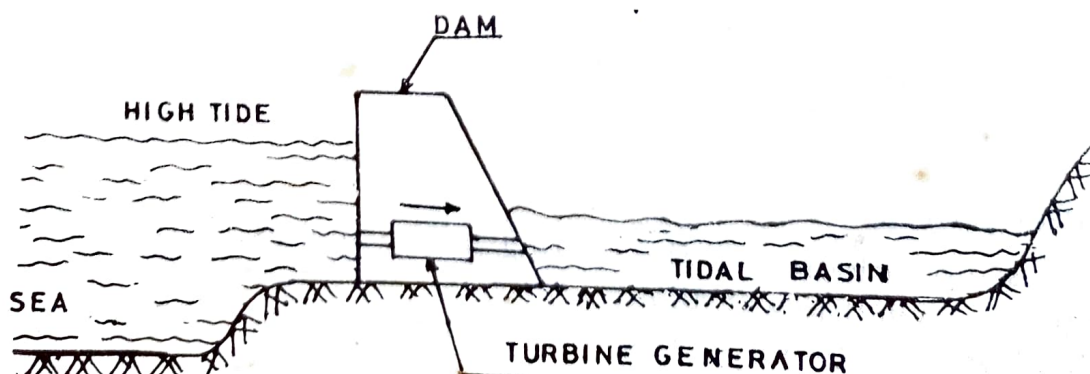
Principle

Tide or wave is periodic rise and fall of water level of the sea. Tides occur due to the attraction of sea water by the moon. Tides contain large amount of potential energy which is used for power generation. When the water is above the mean sea level, it is called flood tide. When the water level is below the mean level it is called ebb tide.

Working

The arrangement of this system is shown in Fig. 2.14. The ocean tides rise and fall and water can be stored during the rise period and it can be discharged during fall. A dam is constructed separating the tidal basin from the sea and a difference in water level is obtained between the basin and sea.

High tide : [Fig. (a)]



(a) High tide

During high tide period, water flows from the sea into the tidal basin through the water turbine. The height of tide is above that of tidal basin. Hence the turbine unit operates and generates power, as it is directly coupled to a generator.

Low tide : [Fig. (b)]

During low tide period, water flows from tidal basin to sea,

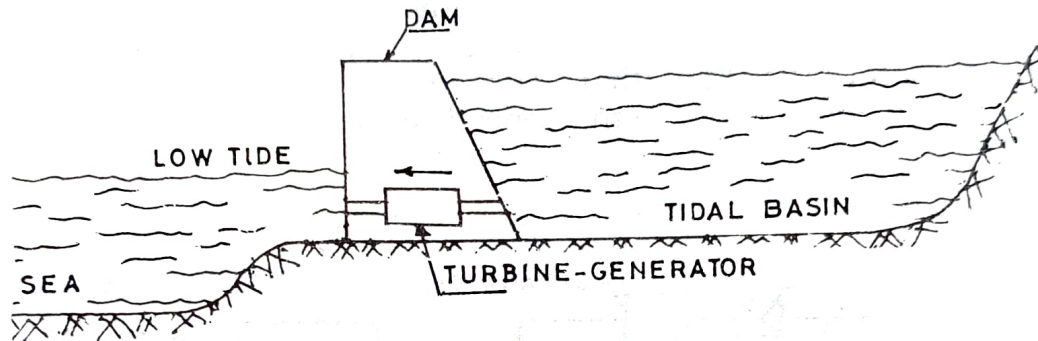


Fig. 214

(b) Low tide

as the water level in the basin is more than that of the tide in the sea. During this period also, the flowing water rotates the turbine and generates power.

The generation of power stops only when the sea level and the tidal basin level are equal. For the generation of power economically using this source of energy requires some minimum tide height and suitable site. Kislaya power plant of 250 MW capacity in Russia and Rance power plant in France are the only examples of this type of power plant.

Advantages of tidal power plants

1. It is free from pollution as it does not use any fuel.
2. It is superior to hydro-power plant as it is totally independent of rain.
3. It improves the possibility of fish farming in the tidal basins and it can provide recreational facilities to visitors and holiday makers.

Disadvantages

1. Tidal power plants can be developed only if natural sites are available on the bay.
2. As the sites are available on the bays which are always far away from load centres, the power generated has to be transmitted to long distances. This increases the transmission cost and transmission losses.

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3. The supply of power is not continuous as it depends upon the timing of tides.
4. The navigation is obstructed.
5. Utilization of tidal energy on small scale is not economical.

2.9 GEO-THERMAL POWER PLANT

It is also a thermal power plant, but the steam required for power generation is available naturally in some part of the earth below the earth surface. According to various theories earth has a molten core. The fact that volcanic action takes place in many places on the surface of earth supports these theories.

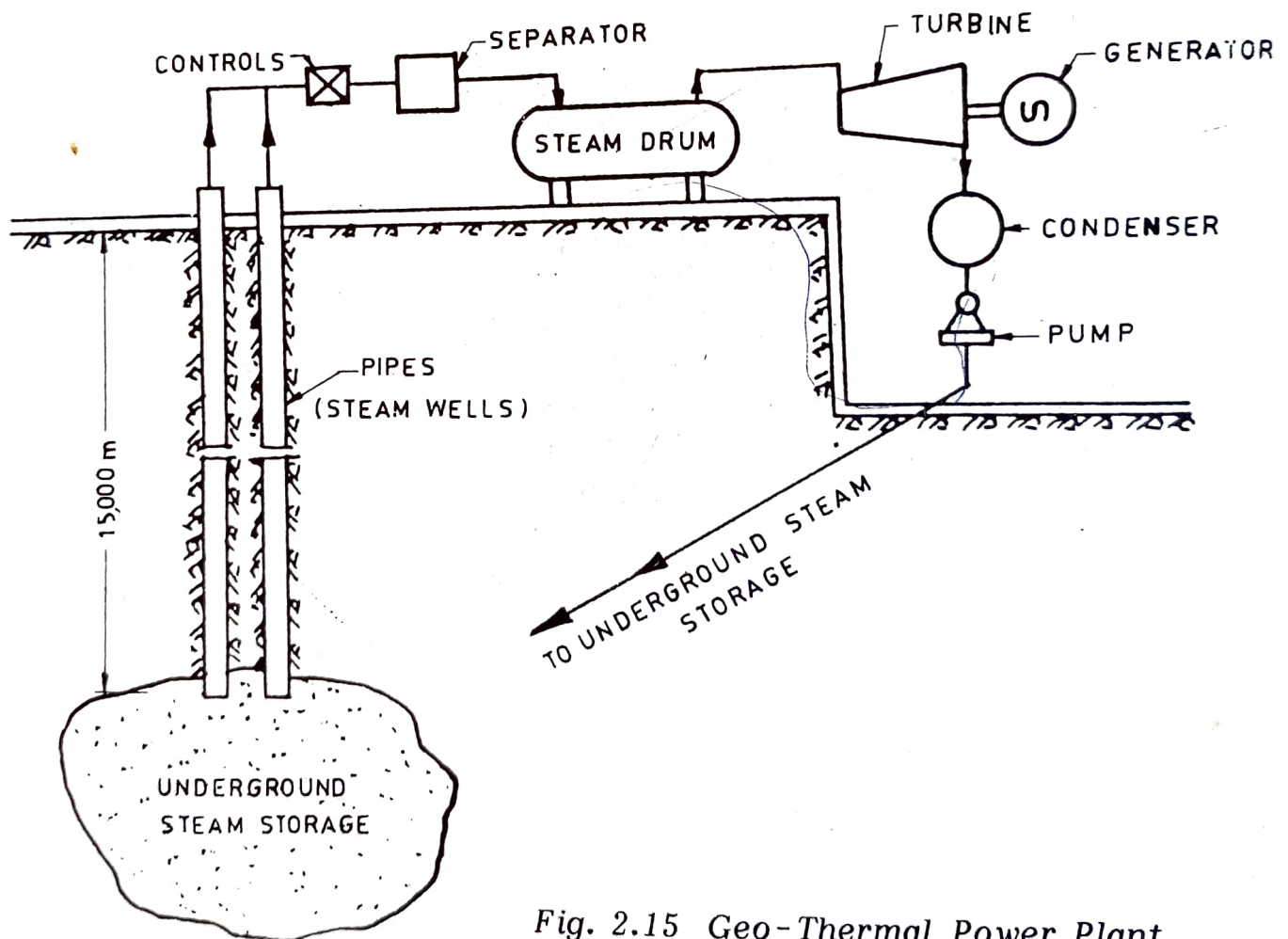


Fig. 2.15 Geo-Thermal Power Plant

Working : Refer Fig. 2.15.

Steam well : Pipes are embedded at places of fresh volcanic action called steam wells, where the molten internal mass of earth

vents to the atmosphere with very high temperatures. By sending water through embedded pipes, steam is raised from the underground steam storage wells to the ground level.

Separator : The steam is then passed through the separator where most of the dirt and sand carried by the steam are removed.

Turbine : The steam from the separator is passed through steam drum and is used to run the turbine which in turn drives the generator. The exhaust steam from the turbine is condensed. The condensate is pumped into the earth to absorb the ground heat again and to get converted into steam.

Location of the plant, installation of equipment like control unit etc., within the source of heat and the cost of drilling deep wells as deep as 15,000 metres are some of the difficulties commonly encountered.

2.12 Factors to be considered for the installation of a Power Plant

- i) Selection of type of plant system
- ii) Site selection
- iii) Selection of prime mover
- iv) Cost

i) Selection of type of plant

The type of power plant to be installed depends upon the available source of energy in a particular region. A thermal power plant should be installed near a coal mine. Hydro-electric power plant should be installed where sufficient head of water and storage capacity are available.

ii) Site selection

The site that is selected should be nearer to the source of fuel and the load of centre. The site should be accessible by road or railway line so that the transportation is easy. The site should be so selected that there is no health hazard or other undesirable consequences to the inhabitants.

iii) Selection of prime mover

The selection of prime movers, viz., steam turbines, hydraulic turbines, diesel engines etc., should be made depending on the load and the type of power plant.

iv) Cost

The cost of a power station should be estimated to be

as low as possible. The cost of transmitting power may be reduced by installing the power station near the load centre. Further the site subsoil conditions should be such that a good foundation can be made at reasonable depth, thereby reducing the cost of the buildings.

2.13 Direct Energy Conversion Systems :

Generally in power plants (thermal, nuclear, solar, etc.), the heat energy is first converted into mechanical energy and then into electrical energy. This double conversion reduces the overall efficiency of the plant (about 30%).

Therefore, great efforts have been made during the last two decades to convert directly heat energy into electrical energy, eliminating intermediate mechanical conversion. This not only reduces the capital cost of the plant and maintenance problems of mechanical parts but also increases the overall efficiency. Such energy conversion systems are known as "direct energy conversion systems".

Thermo-ionic and thermo-electric, photo-voltaic (solar energy is used) and magneto-hydro dynamic (MHD) systems are some of the most promising devices for direct energy conversion. The efficiency of such conversion devices at present is fairly low and capacity is also fairly small. They are not yet available on commercial basis, but these systems hold out great promises and there is a good possibility that such systems may be available by the end of this century.

OCEAN THERMAL ENERGY CONVERSION

Principle

Ocean (sea water) contains a reasonable amount of heat. The surface of the sea water is hot enough, whereas water in a depth of about 600 metres is cold. The temperature difference between the hot surface water and cold water at the depth of 600 m is having a range of approximately 15 to 20° C. The heat energy content of the surface water can be used to heat and boil a low boiling fluid. Its vapour can be made use of, for running a steam turbine.

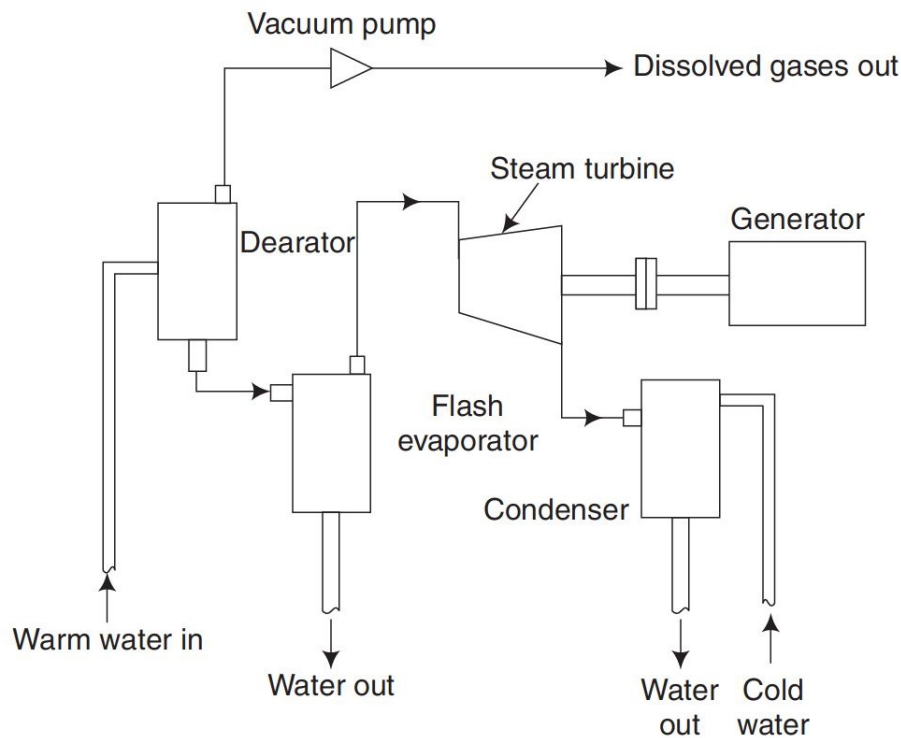
A flash evaporator is used to flash evaporate the hot surface water. The low pressure steam is sent through the turbine, which, in turn, runs a generator.

The Ocean Thermal Energy Conversion (OTEC) system is of two types, viz., Open Cycle System and Closed Cycle System.

Open Cycle OTEC System

The Open cycle uses sea water as the working fluid. The warm water is first sent to the deaerator where the dissolved gases are removed by means of a vacuum pump. Then, the warm water is flash evaporated under a partial vacuum in the flash evaporator. The low pressure steam is then passed through a turbine which extracts energy from it and runs a generator. The vapour, after expansion in the turbine is cooled in the condenser. A spray condenser (direct contact) is used where the condensate is mixed with cooling water and the mixture is discharged into the ocean.

It is to be noted here that since we have to obtain energy from a low pressure steam, extremely large sized turbines must be used.



Schematic diagram of an open-cycle OTEC system

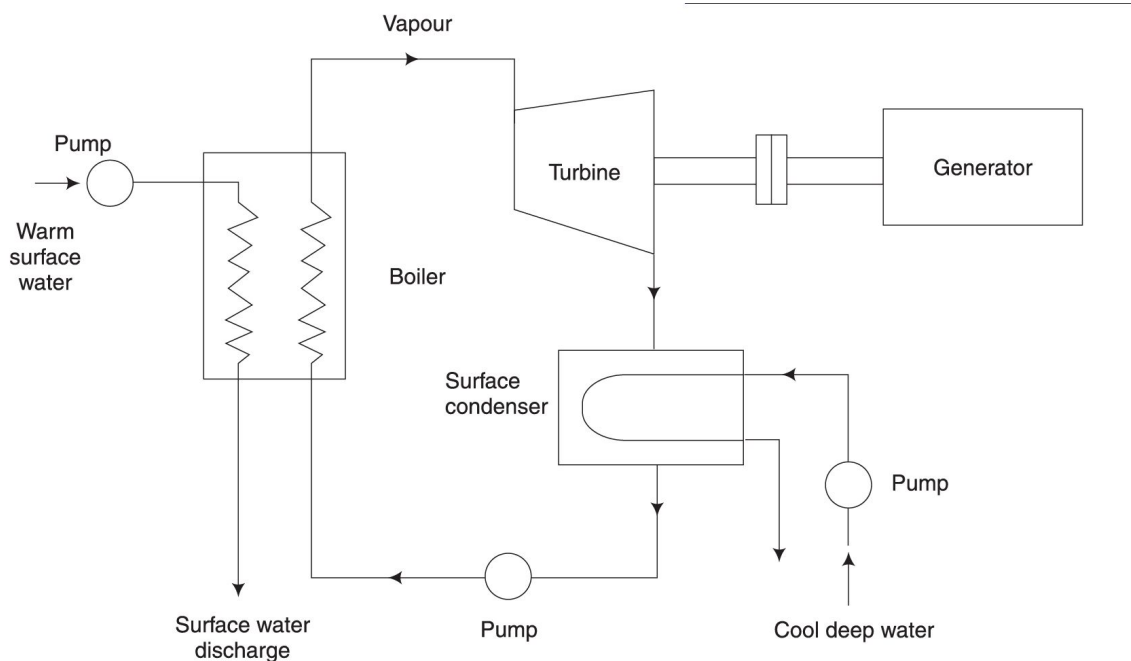
The cost of an open cycle system for producing electric power is significantly greater than that of a closed cycle system. The turbine cost alone amounts to almost half the cost of the entire plant.

Closed Cycle OTEC System

The schematic diagram of a closed cycle OTEC plant is shown in figure. Here the heat exchangers known as evaporators and condensers are the key components of the system. .

In this cycle the heat energy carried by the ocean surface water is given to the secondary working fluids (such as Ammonia, Propene, R-12, etc.,) having low boiling points while operating in a cycle.

The warm surface of sea water carrying some heat energy is made to pass over the heat exchanger namely. evaporator, before it is discharged out to the ocean itself. The heat energy received by the evaporator is imparted to the secondary fluid, which results the secondary fluid to vapourise. The vapourised secondary medium is expanded in the turbine coupled to a generator producing the electrical power.



Schematic of a Closed Cycle OTEC System

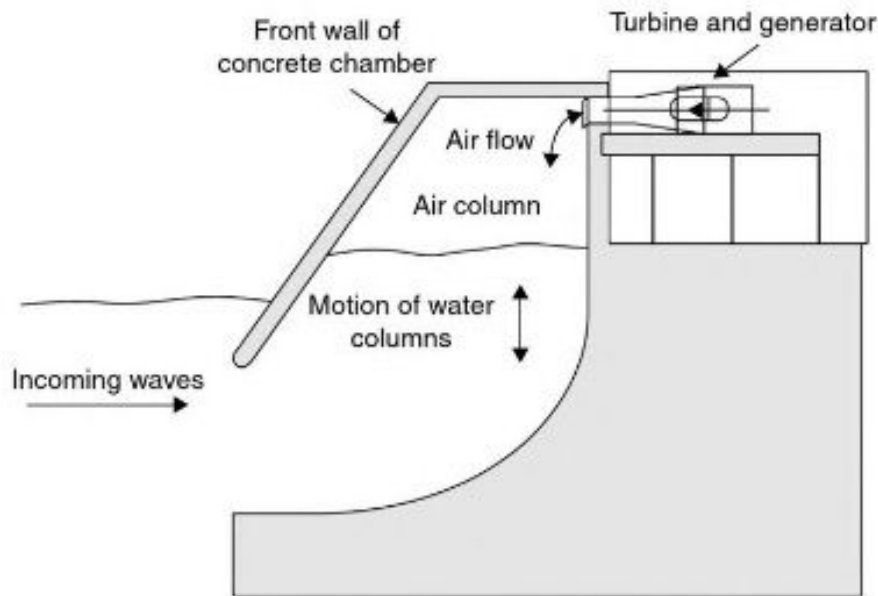
The vapour after expansion is condensed in the condenser by allowing the cold deep sea water to flow over the heat exchanging surface of the condenser. The condensed secondary medium is again sent to the evaporator by means of a pump, and thus the cycle continues. It is roughly estimated that 14% of the gross power is expected to be consumed internally.

WAVE ENERGY

Considering the fact that the waves sweep across the oceans of the world, which cover almost 71% of the earth's surface, the energy available in the waves must be quite abundant. The waves are caused by the interaction of the wind with the waters of the ocean. The wind gets about 1 to 5% of Sun's energy and part of this energy is transferred to the waves. It is estimated that about 10 KW of energy is available for every one meter of wave front. In India, the potential is estimated to be 40,000 MW.

Wave Energy Systems

Many wave extracting systems have been proposed and are being experimented upon. IIT-Chennai, is experimenting on a system using the principle of oscillating water column.



Schematic of an oscillating water column device

Schematic of Wave Power Generation System

The basic principles of operation of this system consists of :

- (i) Causing the wave to enter a chamber and push up a column of air above it.
- (ii) The air getting compressed thus, flows through a small opening, where an air turbine-generator unit is installed.
- (iii) When the wave recedes, it causes suction and the atmospheric air is drawn into the chamber. Thus, there is a regular oscillatory movement of water column causing to and fro motion of the air which in turn, drives the turbine .

The system consists of a caisson which has a hollow chamber inside. The chamber provides space for the wave to rise and fall down enabling an oscillating motion of water to be set up. The air above the water column, on getting pushed up, drives the air turbine.